

The Dragon Chips Back: China's Semiconductor Counter-Offensive and the New Cold War for Silicon Supremacy

Author: Blue Lotus Research Intelligence / CivilisationOS **Date:** August 2026
Classification: Intelligence Brief -- Open Publication

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I. Opening: The Fab That Defies the Embargo

Walk the production floor of a SMIC wafer fabrication facility in Shanghai in August 2026, and you will notice what is absent. There is no ASML extreme ultraviolet lithography machine -- the Dutch device that has become the most strategically important piece of industrial equipment on earth. There is no Synopsys or Cadence software suite running on the engineers' workstations. The etch chambers were not shipped from Lam Research in Fremont, California. The inspection tools are not from KLA in San Jose. What you see instead is a different ecosystem entirely: NAURA deposition furnaces, AMEC etch systems, ACM Research cleaning tools, and process engineers trained at Chinese universities who learned their craft, in many cases, not at foreign graduate schools but at domestic semiconductor programs that have expanded at a pace unmatched since the Soviet era. The chips emerging from this line are not at 3 nanometres. But they work -- and China is making more of them than ever before.

This scene encapsulates the central paradox of the American semiconductor embargo. Conceived as a technological chokehold, the US export control regime that has been progressively tightened from October 2022 through to mid-2026 was designed to do exactly what its architects claimed: arrest China's advance in the technologies that will define military and economic power in the twenty-first century. The logic was compelling in theory. Semiconductor fabrication is the most complex industrial undertaking humanity has ever attempted. It requires the most precise machines ever built, supplied by a narrow oligopoly of Western and Japanese firms that could, in principle, be coordinated to deny China access. Cut off the equipment, the software, and the chips, and China's ambitions -- its AI military applications, its frontier model development, its hypersonic guidance systems -- would be permanently constrained.

The theory has collided with a very different reality. The US semiconductor embargo, rather than stopping China's chip industry, has unleashed what may be the most aggressive

state-directed industrial mobilization since the Manhattan Project. China has responded not with capitulation but with a mobilization of capital, talent, and industrial policy at a scale that has surprised even its sharpest critics. The Big Fund Phase III alone has committed ¥344 billion -- approximately \$47.5 billion -- to domestic semiconductor development [Source: <https://www.tomshardware.com/tech-industry/china-starts-big-fund-iii-spending-usd47-billion-for-ecosystem-and-fab-tools>]. SMIC is expanding at 40,000 twelve-inch-equivalent wafers per month per year [Source: <https://www.digitimes.com/news/a20260212PD212/smic-semiconductor-foundry-demand-infrastructure-2026.html>]. YMTC has claimed the third position in global NAND flash shipments [Source: <https://www.tomshardware.com/tech-industry/ymtc-breaks-into-the-top-three-nand-makers-for-the-first-time>]. CXMT is on trajectory to match Micron's DRAM production capacity by end-2026 [Source: <https://www.tomshardware.com/pc-components/dram/cxmt-close-to-matching-microns-memory-capacity-in-2026-research-claims-would-put-china-on-track-to-become-worlds-second-largest-dram-producer>]. And DeepSeek has demonstrated -- twice over -- that algorithmic efficiency can partially substitute for raw semiconductor horsepower.

The question confronting policymakers, investors, and strategists in Washington, Taipei, Tokyo, Amsterdam, and Singapore is no longer whether China will achieve chip self-sufficiency. It is when, at what cost, and what the geopolitical consequences will be when it does. The answer to that question will shape the architecture of the global economy and the balance of military power for the remainder of this century.

II. The American Embargo -- What Was Cut Off

The architecture of the American semiconductor embargo has been constructed in layers, each tightening the previous one in response to Chinese adaptations. Understanding what precisely has been denied to China -- and what has not -- is essential to assessing the counter-offensive.

The foundational act was the October 2022 Export Administration Regulations (EAR) update, which imposed sweeping new restrictions on the export of advanced computing chips, semiconductor manufacturing equipment, and supercomputing components to China. The controls targeted chips with performance exceeding specific thresholds -- those capable of training large AI models -- as well as the equipment needed to manufacture logic chips below 16 nanometres. This was followed by a series of escalating actions through 2023, 2024, and into 2026 [Source: <https://www.microchipusa.com/industry-news/everything-you-need-to-know-about-the-u-s-semiconductor-restrictions-on-china>].

The specific categories denied to China through these controls are worth enumerating precisely because their scope is often misunderstood in public debate. First, advanced logic chips: foundry access to sub-16nm processes at TSMC, Samsung, and Intel Foundry was effectively severed, meaning Chinese fabless chip designers can no longer send their most advanced designs to leading-edge contract fabs. Second, high-bandwidth memory (HBM): the stacked DRAM packages essential for large-scale AI training, produced exclusively by SK

Hynix, Samsung, and Micron, are controlled for export to China. Third, AI accelerators above defined performance thresholds: Nvidia's A100, H100, H200, and subsequent architectures were restricted; even the downgraded H20 chip -- a version Nvidia designed specifically for the Chinese market at reduced interconnect bandwidth -- was ultimately subject to restrictions that prevented its commercial deployment [Source: <https://semiconductorsinsight.com/us-china-chip-export-controls-h200-2026/>].

Fourth, and most consequentially for China's long-term ambitions, semiconductor manufacturing equipment: 24 categories of advanced chipmaking tools from US companies -- including Applied Materials, Lam Research, and KLA -- now require export licences that are denied in virtually all cases involving Chinese customers operating at advanced nodes [Source: <https://www.microchipusa.com/industry-news/everything-you-need-to-know-about-the-u-s-semiconductor-restrictions-on-china>]. The BIS opened 2026 with an enforcement surge, settling a case against Applied Materials for illegal equipment exports to China through subsidiaries, resulting in a penalty of approximately \$252 million -- the second-highest in BIS history [Source: <https://www.kharon.com/brief/bis-chips-exports-china-enforcement>]. Fifth, EDA software: Synopsys and Cadence design tools for advanced nodes are controlled for export, meaning Chinese chip designers working on cutting-edge projects cannot access the industry-standard software ecosystems.

The allied dimension of the embargo transformed it from a unilateral US action into something approaching a multilateral technological blockade. The Netherlands -- home to ASML, the sole global supplier of extreme ultraviolet lithography machines -- progressively restricted exports of its most advanced tools to China. Initially, Beijing had already been denied EUV access since 2019; the tightening of DUV (deep ultraviolet) lithography controls from 2023 onward extended the restriction to a broader range of ASML's immersion scanner fleet. Japan moved in parallel: in July 2023, Tokyo implemented controls on 23 categories of advanced semiconductor manufacturing equipment, targeting tools capable of producing logic chips at 14 nanometres and below [Source: <https://www.csis.org/analysis/japan-and-netherlands-announce-plans-new-export-controls-semiconductor-equipment>]. A second Japanese round in April 2024 added 21 further items. A third round effective August 2024 extended controls to the entire advanced packaging chain [Source: https://x.com/Teo_Sinamin/status/2084465044096733459].

Yet the embargo has structural limits that its architects acknowledge. The controls apply to new exports; China already had, by 2022, a substantial installed base of equipment purchased in the preceding decade. Chips and tools below specific performance thresholds remain purchasable. Reverse engineering of existing chips -- a practice at which Chinese engineers have demonstrated considerable competence -- is outside the legal reach of US export controls. And the control regime cannot prevent Chinese semiconductor engineers who trained or worked at TSMC, Samsung, or other leading firms from returning to China and applying their knowledge. The talent channel, in particular, has proven extremely difficult to interdict.

III. China's Industrial Response -- The Big Fund and Beyond

The Chinese state's response to the semiconductor embargo has been swift, massive, and strategically coherent in a way that Western commentary has consistently underestimated. The National Integrated Circuit Industry Investment Fund -- universally known as the "Big Fund" (大基金) -- has served as the primary vehicle for state-directed capital mobilization, operating across three phases that represent an escalating commitment of resources.

Phase I, established in 2014, raised approximately RMB 130 billion in capital and invested across 23 companies covering 75 projects [Source: <https://asiasociety.org/policy-institute/chinas-big-fund-30-xis-boldest-gamble-yet-chip-supremacy>]. This initial round focused primarily on domestic foundry build-out, chip design, and packaging -- the most commercially accessible segments of the semiconductor value chain. Phase II, launched in 2019 as US-China technology tensions began to accelerate, raised approximately RMB 200 billion and placed greater emphasis on strengthening the semiconductor supply chain in anticipation of rising restrictions [Source: <https://www.eetimes.com/chinas-big-fund-phase-ii-aims-at-ic-self-sufficiency/>]. Phase III, which began operations on December 31, 2024 and commenced active deployment through 2025-2026, commands a capital base of ¥344 billion -- approximately \$47.5 billion at current exchange rates [Source: <https://www.tomshardware.com/tech-industry/china-starts-big-fund-iii-spending-usd47-billion-for-ecosystem-and-fab-tools>].

The strategic shift encoded in Phase III's investment mandate is significant. Where the earlier phases focused heavily on fab construction and chip design, Phase III has pivoted to the most constrained links in the chain: advanced packaging, semiconductor equipment and materials, and AI computing chips [Source: <https://www.trendforce.com/news/2026/08/07/news-chinas-big-fund-phase-iii-pivot-from-fab-building-to-advanced-packaging-equipment-and-ai-chips/>]. This reflects a sophisticated diagnosis of the embargo's architecture: China already has fabs; what it lacks are the tools to fill them with cutting-edge processes. Phase III's initial investments included ¥93 billion directed at materials producers (ultra-pure chemicals, silicon wafers) and wafer fabrication equipment companies [Source: <https://www.tomshardware.com/tech-industry/china-starts-big-fund-iii-spending-usd47-billion-for-ecosystem-and-fab-tools>].

The Big Fund represents only the most visible portion of China's semiconductor investment architecture. MIIT (Ministry of Industry and Information Technology) subsidies, local government matching funds -- particularly from Shenzhen, Shanghai, Wuhan, and Beijing -- and SOE capital injections multiply the effective total. China has directed semiconductor manufacturers to use at least 50% domestically produced equipment when adding new capacity, creating a guaranteed procurement channel for domestic equipment makers even when their tools are not yet globally competitive [Source: <https://www.techpowerup.com/344595/chinese-government-mandate-local-chipmakers-use-50-percent-domestic-chip-manufacturing-equipment>]. China advanced packaging investment alone saw nearly 10 publicly disclosed projects totalling approximately 400 billion yuan announced between May and July 2026, spanning 2.5D/3D integration, chiplet architecture, fan-out packaging, and automotive chip packaging [Source: <https://pandaily.com/china-advanced-packaging-new-wave-jul2026>].

The geographic architecture of China's semiconductor industrial build-out reflects deliberate policy choices. Shanghai and its surrounding Yangtze River Delta have emerged as the logic foundry heartland, anchored by SMIC's flagship facilities. In January 2026, SMIC moved to acquire full control of its Beijing subsidiary, Semiconductor Manufacturing North China (SMNC), for \$5.79 billion, consolidating its most advanced fabs under unified national direction [Source: <https://enki.ai.com/ai-market-intelligence/smic-ai-chip-strategy-2026-inside-chinas-5nm-power-play/>]. Wuhan hosts YMTC's NAND flash production empire, which has fast-tracked its Phase III fabrication facility for second-half 2026 mass production [Source: <https://www.digitimes.com/news/a20260202PD223/ymtc-nand-flash-fab-production.html>]. Hefei has become China's DRAM capital, hosting two of CXMT's three operating 12-inch fabs [Source: <https://www.tomshardware.com/pc-components/dram/chinas-cxmt-targets-30-percent-dram-memory-market-share-by-2030-with-sixth-mega-fab-future-plans-bottlenecked-by-access-to-advanced-chipmaking-tools>]. Shenzhen hosts Huawei's chip design operations and the newly reported domestic EUV research programme.

The talent mobilization has been equally ambitious, though it has encountered structural constraints. China's semiconductor industry faces a shortfall exceeding 300,000 skilled workers, a gap that SMIC's founder publicly warned of in April 2026 [Source: <https://www.digitimes.com/news/a20260409PD222/china-semiconductor-industry-talent-training-policy.html>]. The country has responded by massively expanding semiconductor engineering programmes -- Tsinghua University has established a specialized chip college -- and by aggressively recruiting overseas Chinese returnees. However, 2026 has brought a setback: the number of high-end overseas semiconductor professionals returning to China fell approximately 45% year-on-year, as tightened US restrictions on the transfer of technology by Chinese-American engineers made the career calculus more perilous [Source: <https://www.suntzurecruit.com/2026/03/25/upgrade-of-semiconductor-talent-war-in-2026-how-can-enterprises-and-job-seekers-break-through/>]. Average annual turnover among high-end semiconductor talent in China runs at over 20%, reflecting acute competition between SMIC, YMTC, CXMT, Huawei, and their equipment-sector counterparts.

Placed in historical context, the scale of China's semiconductor mobilization is without precedent among developing nations. Japan's famous VLSI Project of 1976-1980 -- widely credited with enabling Japan's ascent to semiconductor parity with the United States -- was a coordinated industry-government programme that cost approximately 500 million across its five-year run. Korea's DRAM industry build-out through Samsung and Hynix from the early 1980s drew on government support but was primarily commercially driven. Taiwan's founding of TSMC in 1987 was a single-company, government-backed gambit. China's current semiconductor investment programme, by contrast, involves three-phase committed state capital approaching 100 billion, multiplied by local government co-investment, across every segment of the value chain simultaneously. No historical precedent captures its scope.

IV. The Technology Stack -- What China Can and Cannot Do

The most important analytical task in assessing China's semiconductor counter-offensive is separating the achievable from the aspirational -- the real capabilities that have emerged from the enormous investment programme from the performance claims that serve domestic political and propaganda purposes.

What China Can Do Today

SMIC's current best demonstrated node is its N+3 process, described by independent analysts as 5nm-class in transistor density achieved without EUV, using self-aligned quadruple patterning (SAQP) with existing DUV immersion scanners. SMIC reported that 7nm-class capacity is being heavily prioritized for domestic customers, particularly Huawei, and that yields at the N+2/N+3 nodes are reported in the range of 20-40% -- significantly below TSMC's mature-node yields of 90%+ but commercially viable under a protected domestic market and state subsidy structure [Source: <https://chinamade.tech/blog/smic-explained>]. SMIC guided an addition of approximately 40,000 twelve-inch-equivalent wafers per month by end-2026 on top of the roughly 50,000 added the prior year [Source: <https://www.digitimes.com/news/a20260212PD212/smic-semiconductor-foundry-demand-infrastructure-2026.html>].

Huawei's Kirin 2026 chip, set to debut in the Mate 90 series in autumn 2026, represents a genuine engineering achievement in design-level innovation. Rather than relying on advancing to a smaller process node -- impossible given SMIC's equipment constraints -- Huawei claims to have achieved a 55% increase in transistor density over the prior generation Kirin 9030 Pro by adopting dual-layer LogicFolding, delivering a density of 238 million transistors per square millimetre -- a figure that, the company asserts, rivals TSMC's 3nm transistor density [Source: <https://semiwiki.com/forum/threads/huawei-plans-1-4-nm-chips-by-2031-kirin-2026-chip-238-mtr-mm2-transistor-density-rivaling-tsmc%E2%80%99s-3nm.25154/>]. Power consumption is cut by 41% versus the prior generation. Huawei has published research on hybrid bonding for 3D chip stacking, pursuing chiplet integration as a pathway around node limitations [Source: <https://wccftech.com/huawei-kirin-2026-paper-shows-hybrid-bonding-for-3d-stacking-to-boost-competition/>].

CXMT -- Chang Xin Memory Technologies -- has executed what may be the most startling capacity expansion in Chinese semiconductor history. Operating three 12-inch DRAM fabs (two in Hefei, one in Beijing) with combined monthly capacity of approximately 300,000 wafers, CXMT is projected to exit 2026 at 350,000 wafers per month, closing to within 25,000 wafers per month of Micron's targeted capacity [Source: <https://www.tomshardware.com/pc-components/dram/cxmt-close-to-matching-microns-memory-capacity-in-2026-research-claims-would-put-china-on-track-to-become-worlds-second-largest-dram-producer>]. The company is producing 24Gb modules at approximately 6,000 MT/s and LPDDR5X-10667 in 12Gb and 16Gb capacities in mass production [Source: <https://www.tweaktown.com/news/112680/chinas-cxmt-is-on-track-to-nearly-match-microns-dram-production-capacity-by-the-end-of-2026/index.html>].

Technology-wise, CXMT relies on D1a/D1b/D1c DRAM generations using DUV multi-patterning rather than EUV -- a technically inferior but functional approach. CXMT plans

a sixth mega-fab targeting 30% global DRAM market share by 2030 [Source: <https://www.toms hardware.com/pc-components/dram/chinas-cxmt-targets-30-percent-dram-memory-market-share-by-2030-with-sixth-mega-fab-future-plans-bottlenecked-by-access-to-advanced-chipmaking-tools>].

YMTC's Xtacking architecture NAND flash programme has achieved what would have seemed impossible when the company was placed on the Entity List in December 2022. As of second quarter 2026, YMTC held approximately 14% of global NAND flash shipments, placing it third globally ahead of Kioxia -- the first time a Chinese memory company has achieved top-three status in any major semiconductor product category [Source: <https://www.tomshardware.com/tech-industry/ymtc-breaks-into-the-top-three-nand-makers-for-the-first-time>]. YMTC is mass-producing 294-layer 3D NAND with yields reported above 90%, with development underway on devices exceeding 300 layers [Source: <https://www.tweaktown.com/news/113106/chinas-ymtc-has-climbed-to-third-place-in-global-nand-flash-shipments/index.html>]. Combined production capacity across its two Wuhan facilities stands at approximately 200,000 wafers per month, with a Phase III facility fast-tracked for second-half 2026 and two further fabs in planning [Source: <https://www.digitimes.com/news/a20260202PD223/ymtc-nand-flash-fab-production.html>].

China's domestic equipment sector has made measurable progress. NAURA, AMEC, and ACM Research Shanghai -- the "Big Three" of Chinese semiconductor equipment -- collectively hold approximately 45-50% combined domestic market share [Source: <https://greathandshake.com/en/chinas-semiconductor-equipment-industry-how-naura-amec-and-domestic-toolmakers-are-closing-the-gap/>]. NAURA's oxidation and diffusion furnaces account for more than 60% of equipment on SMIC's 28nm production lines; NAURA's 2026 revenue is expected to exceed RMB 50 billion, making it China's largest semiconductor equipment company [Source: <https://www.trendforce.com/news/2026/01/12/news-chinas-domestic-chip-equipment-adoption-beats-2025-target-at-35-led-by-naura-amec/>]. AMEC's 5nm-class etch tool has entered validation on TSMC production lines, a milestone that signals credible international competitiveness even if the tool is not yet deployed at leading-edge nodes in China [Source: <https://www.trendforce.com/news/2026/01/12/news-chinas-domestic-chip-equipment-adoption-beats-2025-target-at-35-led-by-naura-amec/>]. ACMR's single-wafer cleaning tools have secured orders from Hua Hong's 28nm lines. The domestic equipment adoption rate exceeded the 2025 target of 35%, with etch and deposition substitution surpassing 40% [Source: <https://www.trendforce.com/news/2026/01/12/news-chinas-domestic-chip-equipment-adoption-beats-2025-target-at-35-led-by-naura-amec/>].

In EDA software, Empyrean Technology remains the principal domestic alternative to Synopsys and Cadence. The company achieved RMB 1.325 billion in revenue in 2025 and has launched China's first full-process EDA solution for memory chip design, covering Flash and DRAM from design through verification to manufacturing sign-off. Its analog EDA tools have received international certification at multiple advanced process nodes [Source: <https://www.trendforce.com/news/2025/08/19/news-empyrean-reportedly-unveils-chinas-first-full-process-eda-platform-for-memory-chip-production/>]. However, Empyrean does not yet offer a complete

integrated suite for cutting-edge logic design at GAA transistor nodes (3nm and below), and its point-solution approach incurs significant productivity penalties versus the integrated workflows of Western tools.

The Three Hard Ceilings

Despite the genuine progress documented above, three structural limitations define the ceiling of China's current semiconductor capability -- and each is extraordinarily difficult to breach.

The first and hardest ceiling is EUV lithography. ASML is the sole global supplier of extreme ultraviolet lithography machines, and a single EUV scanner requires components sourced from approximately 30 countries, assembled through supply chains built over three decades that cannot be replicated at any realistic speed or cost [Source: <https://thediplomat.com/2026/07/chinas-euv-lithography-progress-parsing-signal-from-noise/>]. China has reportedly constructed a prototype EUV device in a high-security Shenzhen laboratory -- described as operational and capable of generating EUV light at 13.5nm wavelength -- but it achieves only 100-150 watts of output power versus ASML's 250-watt capability in 2017 (which enabled 125 wafers per hour) and current standard of 600 watts [Source: <https://asiatimes.com/2026/07/chinas-duv-lithography-still-lags-asml-by-four-generations/>]. The Chinese laser system loses too much energy to heat and non-EUV wavelengths rather than usable 13.5nm light, making wafer throughput commercially uncompetitive by multiple orders of magnitude. Even SMEE's DUV scanner programme -- China's nearest-term lithography milestone -- has only reached mass production at 90nm with a 28nm immersion model in development [Source: <https://asiatimes.com/2026/07/chinas-duv-lithography-still-lags-asml-by-four-generations/>]. Goldman Sachs analysis published in August 2026 estimated that China's chip gap with the frontier will narrow to approximately 34% by 2035, with lithography remaining the binding constraint throughout [Source: <https://www.techtimes.com/articles/325589/20260826/goldman-sees-china-chip-gap-narrowing-34-2035-lithography-still-binding-limit.htm>].

The second ceiling is sub-20nm DRAM design. The process innovations that enabled Samsung, SK Hynix, and Micron to manufacture DRAM at sub-20nm geometries were accumulated over decades of proprietary development -- a body of tacit manufacturing knowledge that cannot be transferred through published papers or reverse engineering alone. CXMT's impressive capacity numbers mask a technology generation gap: its current production nodes are estimated to be 2-3 process generations behind the leading Samsung and SK Hynix D1c/D1d nodes. For commodity DRAM this gap is commercially manageable under state subsidy; for HBM memory needed in AI training clusters, it is insurmountable at current technology levels.

The third ceiling is the EDA software ecosystem. Synopsys and Cadence collectively command over 90% of the advanced node electronic design automation market, and their deepest moat is not the software per se but the foundry Process Design Kits (PDKs) -- proprietary libraries of device models, design rules, and verified cell libraries developed in close

collaboration with leading-edge fabs. Empyrean has made real progress in memory and analog design; it cannot yet offer a competitive path for designing leading-edge logic at sub-5nm nodes. The productivity penalty is real and severe.

The Workarounds

Understanding China's semiconductor strategy requires recognising that these ceilings are being worked around rather than broken through, with three primary technical approaches.

DUV multi-patterning is the foundational workaround. SMIC employs self-aligned quadruple patterning (SAQP) with existing ASML DUV immersion scanners -- a technique that involves exposing each chip layer multiple times with slightly offset patterns to achieve feature sizes below the theoretical resolution limit of the lithography tool. Multi-patterning roughly quadruples the number of lithography steps, increasing mask complexity and compounding positional error through overlay accumulation across passes [Source: <https://drrobertcastellano.substack.com/p/how-china-is-reaching-5nm-without>]. Each additional patterning pass introduces another opportunity for misalignment, contamination, or equipment failure. The result is structurally yield-challenged production -- the 20-40% yield range at SMIC's most advanced nodes versus 90%+ at TSMC reflects this compounding cost. But at subsidized prices in a protected domestic market, yields of 20% can sustain a commercial operation.

Chiplet architecture and advanced packaging represent the more elegant and potentially more durable workaround. Rather than manufacturing an entire advanced logic chip on a single die -- which requires the most demanding lithography -- China is investing aggressively in disaggregated design: breaking complex chips into smaller dies, each manufacturable at available nodes, and assembling them through sophisticated packaging techniques (fan-out, 2.5D silicon interposer, 3D stacking with hybrid bonding). SMIC established an advanced packaging research institute in Shanghai in 2026, pursuing 2.5D/3D integration and system-in-package technologies [Source: <https://www.packnode.org/en/innovation/smic-advanced-packaging-research-institute-shanghai-2026>]. Huawei's Kirin 2026 chiplet research explicitly invokes hybrid bonding as a 3D stacking pathway [Source: <https://wccfttech.com/huawei-kirin-2026-paper-show-s-hybrid-bonding-for-3d-stacking-to-boost-competition/>].

The third workaround is domain-specific ASIC design: engineering chips optimised for specific applications rather than general-purpose computing. Huawei's Ascend 910 series represents this approach -- custom silicon designed specifically for AI inference workloads, where the performance gap against Nvidia's general-purpose accelerators is narrower than in training. Each Ascend 910C delivers approximately 60% of an Nvidia H100's inference performance while running on SMIC's available nodes [Source: <https://www.csis.org/analysis/deepseek-huawei-export-controls-and-future-us-china-ai-race>].

V. The ASEAN Equipment Rerouting -- The Grey Channel

No analysis of China's semiconductor counter-offensive is complete without examining the systematic flow of equipment through Southeast Asian intermediaries that has partially compensated for the direct equipment restrictions. The numbers here are stark and draw from primary trade data.

In 2025, China's imports of semiconductor equipment from Singapore reached *5.7 billion -- a year-on-year increase of more than 173.4 billion*, more than doubling from 2024 levels [Source: <https://eastasiaforum.org/2026/06/20/chinas-chipmaking-supply-chain-runs-through-southeast-asia/>]. Over the same period, China's direct semiconductor equipment imports from the United States fell by more than 34% to approximately \$2 billion, the lowest level since 2017 [Source: <https://www.trendforce.com/news/2026/04/15/news-china-reportedly-sees-record-2025-chip-tool-imports-from-singapore-malaysia-u-s-imports-hit-lowest-since-2017/>]. The arithmetic is suggestive: as direct US-origin equipment dried up, ASEAN-origin equipment swelled by a commensurate amount.

The mechanism operates at multiple levels of legality and opacity. Equipment sold by Western manufacturers to ASEAN-based entities -- semiconductor companies operating in Singapore or Malaysia, equipment distributors, trading companies -- can be legally re-exported to China provided the equipment is classified below the control thresholds that require export licences. Many categories of legacy DUV lithography tools, deposition systems, and etch equipment fall below these thresholds precisely because the control architecture was designed to target leading-edge capability rather than established production nodes. Where equipment above the threshold is involved, violations of US, Dutch, or Japanese export control law are alleged but difficult to prove without access to end-user documentation. BIS has limited investigative resources and the evidentiary chain across multi-jurisdiction transactions is complex to reconstruct [Source: <https://itif.org/publications/2026/05/26/asean-becomes-middleman-in-us-china-tech-trade/>].

The context for this grey channel is the broader semiconductor investment landscape in ASEAN. Since 2020, the region has attracted approximately \$60.8 billion in semiconductor foreign direct investment, with nearly 80% concentrated in Singapore and Malaysia [Source: <https://www.sourceofasia.com/southeast-asia-semiconductor-2025-2026/>]. Singapore functions as the central export hub for both high-value production and intermediate goods. The country simultaneously hosts GlobalFoundries, Micron's backend operations, and a large cohort of equipment and materials firms closely tied to US technology and compliance systems. Malaysia, particularly in Penang, hosts an established semiconductor packaging and testing ecosystem that has been among the fastest-growing nodes in the global chip supply chain.

This dual dependency creates the ASEAN semiconductor dilemma in its sharpest form. Singapore and Malaysia cannot unilaterally foreclose equipment flows to China without severing commercial relationships with the Chinese semiconductor industry that represent substantial revenue for ASEAN-based distributors, logistics providers, and trading intermediaries. At the same time, continuing to facilitate such flows attracts escalating US pressure under the Chip 4

Alliance framework and risks secondary sanctions if violations of US export law are proven.

Japan's progressive tightening of its equipment export control regime has narrowed the available pool. The three rounds of Japanese equipment controls implemented in July 2023, April 2024, and August 2024 now cover 23 categories of advanced tools across the entire semiconductor manufacturing sequence, including advanced packaging equipment [Source: <https://csis.org/analysis/japan-and-netherlands-announce-plans-new-export-controls-semiconductor-equipment>]. These controls effectively require Japanese equipment makers -- Tokyo Electron, Shin-Etsu, Sumco -- to obtain export licences that are denied at extremely high rates for China-destined shipments. The practical effect has been to redirect Japanese equipment via Korean, Taiwanese, or ASEAN intermediaries, raising the cost and complexity of procurement without eliminating it.

Net assessment: the ASEAN rerouting channel has been materially significant for China's legacy-node fab expansion -- particularly for the 28nm-and-above production that dominates CXMT's and YMTC's current capacity build. It has been inadequate as a channel for leading-edge equipment, where the tools involved are large, expensive, individually identifiable, and closely monitored by their manufacturers. China's most advanced fab expansion has had to proceed with equipment it already possessed, creatively extended through multi-patterning techniques, rather than freshly imported leading-edge systems.

VI. DeepSeek and the AI Chip Bypass

The launch of DeepSeek's R1 model in January 2026 sent a shock through the global technology industry that has not fully dissipated. The model achieved performance broadly comparable to OpenAI's GPT-4 class on standard benchmarks while reportedly requiring a fraction of the training compute, and was trained in part on Nvidia H800 chips -- a downgraded, lower-interconnect export version that China was permitted to receive before controls were tightened -- alongside Huawei Ascend accelerators [Source: <https://www.csis.org/analysis/deepseek-huawei-export-controls-and-future-us-china-ai-race>].

The implications extend well beyond the headline performance comparison. DeepSeek demonstrated that the relationship between training compute and model capability is not fixed -- that algorithmic innovations in architecture, training efficiency, and inference optimization can substitute partially for raw silicon. This is precisely the kind of substitution that an export control regime cannot block: ideas do not appear on any entity list. Moreover, DeepSeek V4, launched in April 2026, was the first frontier-class AI model built and deployed entirely on Chinese domestic semiconductor infrastructure, running on Huawei Ascend 950PR chips [Source: <https://fortune.com/2026/04/24/deepseek-v4-ai-model-price-performance-china-open-source/>]. The symbolic and practical significance of this milestone -- China running a GPT-4-competitive model on domestically designed and manufactured chips -- cannot be overstated.

Huawei's Ascend 910C, which has emerged as the primary domestic alternative to Nvidia hardware in China, delivers approximately 60% of an H100's inference performance [Source:

<https://www.csis.org/analysis/deepseek-huawei-export-controls-and-future-us-china-ai-race>].

For inferencing AI models -- running trained models to generate outputs -- this performance level is adequate for a wide range of commercial and industrial applications. Barclays estimated that by 2026, approximately 70% of AI compute demand consists of inference rather than training, which materially reduces the strategic importance of the training hardware gap [Source: <https://lushbinary.com/blog/deepseek-v4-huawei-ascend-ai-infrastructure-strategy/>]. Huawei captured roughly half of China's \$50 billion domestic AI chip market by positioning Ascend as the primary alternative to Nvidia, which has seen its China market share crater toward zero following the progressive tightening of controls and China's own retaliatory restriction on H200 imports [Source: <https://www.tomshardware.com/tech-industry/huawei-expects-12-billion-in-ai-chip-revenue-this-year-as-nvidias-china-market-share-hits-zero>].

The algorithmic efficiency race that the chip embargo has inadvertently incentivized represents the deepest strategic challenge for the US approach. Constraining Chinese access to hardware has created the most powerful possible incentive structure for Chinese researchers and engineers to innovate in software and algorithmic approaches -- to squeeze more capability out of constrained silicon. The gap between Chinese and US AI development may therefore close faster in the inference domain, where algorithmic efficiency matters most, than the hardware gap would suggest.

This dynamic has a specific application to China's national AI laboratory structure. Organisations including the Beijing Academy of Artificial Intelligence (BAAI), Zhipu AI, Moonshot AI (developers of Kimi), and DeepSeek itself are navigating the hardware constraint through a combination of efficient architecture design, model distillation, and inference optimization -- approaches that are arguably producing research contributions of global significance precisely because the hardware constraint forces creative solutions. The strategic irony is that the embargo, by denying China access to abundant compute, may be producing a more compute-efficient Chinese AI research tradition.

VII. The Geopolitical Endgame -- Three Scenarios for 2030

Scenario analysis of China's semiconductor trajectory must be grounded in the technical constraints identified above while remaining alert to the possibility of nonlinear developments -- breakthroughs, policy shifts, or market adaptations -- that could substantially alter the trajectory. Three scenarios bracket the plausible range.

Scenario 1: Partial Self-Sufficiency (Most Likely; assessed probability ~60%)

China achieves dominant self-sufficiency in mature nodes at 28nm and above by 2028-2029. At these nodes, domestic equipment (NAURA, AMEC, ACMR), domestic lithography (SMEE DUV), and domestic EDA (Empyrean) collectively cover the required process steps, meeting the government's mandate of 50% domestic equipment content. CXMT achieves 30% global DRAM market share by 2030 through volume, though remaining 2-3 generations behind Samsung/SK Hynix technology. YMTC sustains its top-three NAND position and potentially challenges

Kioxia for second place.

At advanced nodes (7-14nm), China achieves adequate manufacturing capability via SMIC multi-patterning with continued ASEAN equipment procurement maintaining partial legacy DUV access. Yields remain structurally below TSMC but are viable under domestic demand guarantee and state subsidy. The 80% self-sufficiency target by 2030 cited by Chinese industry leaders [Source: <https://www.newelectronics.co.uk/content/blogs/china-reportedly-targets-80-chip-self-sufficiency-by-2030>] remains aspirational for overall chip content but may be achievable for specific categories -- power semiconductors, display drivers, Wi-Fi chips, and consumer-grade MCUs -- while sub-5nm logic and HBM remain points of dependency.

For TSMC and ASML, this scenario implies continued and possibly growing demand as the non-China world accelerates technology investment to maintain the lead. Nvidia recovers China revenue through allowed sales channels at whatever threshold the US government maintains, while Huawei's Ascend dominates the Chinese AI market. The JS-SEZ and ASEAN semiconductor ecosystem thrive as the neutral manufacturing zone between bifurcated US-China supply chains.

Scenario 2: Technological Breakthrough (Less Likely; assessed probability ~20%)

China achieves one of two possible breakthroughs that substantially resets the strategic calculus. Either SMIC's multi-patterning programme achieves 5nm-equivalent yields sufficient for high-volume commercial production -- a genuine engineering breakthrough that would validate the DUV-plus-patterning approach at economic scale -- or China's domestic EUV programme achieves a functional prototype with throughput sufficient for pilot production. Huawei's roadmap targets 1.4nm-equivalent chip capability by 2031 [Source: <https://winbuzzer.com/2026/05/26/huawei-targets-14a-equivalent-kirin-chips-by-2031-xcxwn/>]; realising any significant portion of this trajectory would represent a major strategic shift.

This scenario would trigger an emergency response from the US-led alliance: comprehensive extraterritorial controls, accelerated sanctions on Chinese equipment companies, and probably direct diplomatic pressure on ASEAN governments to close the grey channel completely. ASML would face pressure to remotely disable or deactivate tools previously sold to Chinese customers. The global semiconductor industry would bifurcate more rapidly than in Scenario 1.

Scenario 3: Technology War Escalation (Least Likely for full decoupling but increasing risk; assessed probability ~20%)

The current temporary suspension of China's rare earth export controls -- the package of MOFCOM Announcements 55-58, 61, and 62 suspended through November 10, 2026 -- expires, and both sides elect to allow the escalation to resume [Source: <https://carraglobe.com/china-rare-earth-export-controls-2026/>]. China reimplements full rare earth and magnet export controls simultaneously with the US imposing a blanket ban on all semiconductor equipment exports to China (including legacy DUV) -- a move that bipartisan lawmakers called for in early 2026 [Source: <https://www.csis.org/analysis/limits-chip-export-controls-meeting-china-challenge>]. ASEAN

governments, under US pressure, fully enforce equipment re-export controls. The global semiconductor industry bifurcates into two largely separate ecosystems, each with its own supply chain, standards body, and customer base.

Under full bifurcation, two distinct global semiconductor value chains emerge: a US-led ecosystem anchored by TSMC (Taiwan/Arizona), Samsung (Korea), ASML (Netherlands), and the major US equipment and EDA firms, serving North America, Europe, Japan, South Korea, Taiwan, and allied ASEAN states; and a China-led ecosystem anchored by SMIC, CXMT, YMTC, NAURA, AMEC, and Empyrean, serving China's domestic market and potentially select non-aligned nations seeking diversified semiconductor supply. For TSMC and ASML, bifurcation actually reduces complexity; for the global economy, it implies duplicative investment and structural cost inflation in both chips and everything that uses them.

VIII. Implications for Singapore and the JS-SEZ

Singapore occupies a position of peculiar strategic delicacy in the semiconductor cold war. The city-state simultaneously hosts US-aligned semiconductor companies -- GlobalFoundries' 12-inch fab, Micron's extensive backend operations, and a large contingent of equipment and materials firms deeply embedded in US technology and compliance regimes -- while serving as the largest single re-export hub for semiconductor equipment flowing toward China. Both revenue streams are material; neither can be lightly surrendered.

The Johor-Singapore SEZ (JS-SEZ), formally launched in March 2026, has positioned itself as an advanced manufacturing platform precisely suited to the gap that embargo-driven supply chain restructuring is creating [Source: <https://www.mida.gov.my/media-release/mida-anchors-malaysia-as-a-strategic-semiconductor-supply-chain-partner-under-johor-singapore-sez/>]. The SEZ combines Malaysia's land-intensive manufacturing capacity and lower cost structure with Singapore's regulatory credibility, connectivity, and financial infrastructure. By the third quarter of 2025, Johor had recorded approved investments of approximately US\$23 billion, significantly concentrated in advanced manufacturing and digital infrastructure [Source: <https://www.lundgreensinvestorinsights.com/building-up-the-johor-singapore-special-economic-zone/>]. A China-based chip company was reported in January 2026 to be close to signing an anchor investment deal in the JS-SEZ [Source: <https://www.nst.com.my/business/economy/2026/01/1357215/china-based-chip-giant-set-anchor-js-sez-signing-deal-soon>].

The JS-SEZ semiconductor strategy is correctly positioned in the 28nm-and-above mature node space, where Chinese competition is real but manageable -- mature nodes are globally oversupplied and Chinese expansion is primarily displacing other incumbents rather than establishing a technological frontier. Front-end advanced fab operations, which would require leading-edge EUV lithography and direct engagement with the most politically sensitive equipment flows, remain outside the SEZ's near-term scope. The natural competitive advantage of the JS-SEZ is in OSAT (outsourced semiconductor assembly and testing), backend packaging, and the supply chain services that support both US-aligned and China-aligned customers through

the geographic and regulatory arbitrage the dual-country structure provides.

The central risk for the JS-SEZ semiconductor thesis is the collapse of the ASEAN neutral hub position under escalating US pressure. Singapore's position as a re-export hub for semiconductor equipment -- the \$5.7 billion in China-bound flows documented for 2025 -- is legally precarious. BIS enforcement actions and allied government pressure to implement stricter end-user verification requirements could force Singapore to choose between its US relationship and its China-facing trade role. If Singapore and Malaysia were compelled to fully enforce export controls, the ASEAN neutral hub thesis would collapse and with it a significant portion of the economic rationale for the JS-SEZ's semiconductor positioning. The more robust investment thesis under this risk scenario favours OSAT plays -- companies like Inari Amertron and Unisem -- that derive their value from technical capability and customer relationships rather than from regulatory arbitrage.

IX. Superforecast Table

The following forward projections are based on web-sourced data points as cited and represent analytical best estimates under Scenario 1 (Partial Self-Sufficiency). Data pending notation where live sources were insufficient.

Metric	2026 (Now)	2028	2030
China's best demonstrated logic node	~5nm (N+3, low yield, DUV SAQP)	~5nm (improved yield); 3nm EUV: data pending	~3nm (if domestic EUV matures); otherwise 5nm-class
SMIC wafer capacity (12-inch equiv., 000 WSPM)	~300,000	data pending	data pending
China DRAM self-sufficiency (volume share)	~10-15% (CXMT ramp)	~20-25%	~30% (CXMT 2030 target) [Source: https://www.tomshardware.com/pc-components/dram/chinas-cxmt-targets-30-percent-dram-memory-market-share-by-2030]
China NAND self-sufficiency (global share)	~14% (YMTC) [Source: https://www.tomshardware.com/tech-industry/ymtc-breaks-into-the-top-three-nand-makers-for-the-first-time]	~20%	~25-30%
China domestic equipment share (of domestic spend)	~35% (beat 2025 target) [Source: https://www.trendforce.com/news/2026/01/12/news-chinas-domestic-chip-equipment-adoption-beats-2025-target-at-35-led-by-naura-amec/]	~50% (mandate threshold)	~60%

Metric	2026 (Now)	2028	2030
China overall semiconductor self-sufficiency	~33% [Source: https://asia.nikkei.com/business/tech/semiconductors/china-chip-sector-targets-80-self-sufficiency-with-us-in-its-sights]	~50%	80% (government target; realistically ~60%)
Big Fund cumulative investment	~\$100B (Phases I-III combined) [Source: https://www.tomshardware.com/tech-industry/china-starts-big-fund-iii-spending-usd47-billion-for-ecosystem-and-fab-tools]	~\$130B	data pending
Huawei Ascend AI chip market share (China)	~50% of \$50B market [Source: https://www.tomshardware.com/tech-industry/huawei-expects-12-billion-in-ai-chip-revenue-this-year-as-nvidias-china-market-share-hits-zero]	data pending	data pending
China semiconductor talent shortage	>300,000 [Source: https://www.digitimes.com/news/a20260409PD222/china-semiconductor-industry-talent-training-policy.html]	data pending	data pending

X. Data Sources Register

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